

VetRad-ForensicsNet: Toward a Benchmark for Veterinary Radiograph Forensics Against AI-Generated Fraud

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Abstract—Generative models can now synthesize medical images realistic enough to deceive both human experts and automated systems, a risk recently confirmed for human radiographs by a 2026 *Radiology* study in which board-certified radiologists misclassified a meaningful fraction of AI-generated chest X-rays as authentic. This vulnerability directly threatens insurance underwriting, where fraudulent diagnostic imagery can be used to inflate or fabricate claims. While medical deepfake detection has received growing attention in human radiology, no prior published dataset or method addresses *veterinary* diagnostic imaging, despite documented and costly fraud in the pet insurance industry. This paper reframes an initial proof-of-concept dual-stream detector as the seed of a broader benchmark, VetRad-ForensicsNet, for veterinary radiograph forensics. We report an internal audit that uncovered a resolution-history confound capable of producing a spuriously perfect detector on its own, right of any generator-specific forensic signal, and confirm it empirically (Section VI-A). We identify a second, substantially larger openly licensed dataset (VetXRay, 9,882 canine and feline thoracic radiographs, CC BY 4.0) beyond the single 152-image Mendeley collection used in the initial proof of concept, and lay out a multi-generator, cross-generalization, robustness, and explainability evaluation protocol. **[EXPERIMENT REQUIRED: completar con los resultados corregidos una vez determinados: accuracy, AUC, IC 95% del detector sobre el holdout real, y el resumen de la matriz cross-generator]** We release the dataset, code, and trained weights publicly to seed further research in this unexplored niche.

Index Terms—veterinary radiology, medical deepfake detection, generative adversarial networks, diffusion models, insurance fraud, frequency-domain forensics, cross-generator generalization, synthetic medical imaging

I. INTRODUCTION

Generative adversarial networks (GANs) and diffusion models have reached a level of visual fidelity that makes synthetic medical images increasingly difficult to distinguish from real acquisitions. Tordjman et al., in a multi-institutional study published in *Radiology* in March 2026, exposed 17 radiologists from 12 institutions across six countries to a balanced set of 264 authentic and AI-generated chest radiographs (produced with tools including ChatGPT-based generators and Stanford’s RoentGen model); reader accuracy ranged from 58% to 92%, meaning a meaningful fraction of fabricated images escaped detection entirely, and the same vulnerability extended to automated multimodal models [1]. The study explicitly names insurance fraud, litigation, and factitious disorder (Munchausen syndrome) as realistic misuse scenarios [1].

This risk is not hypothetical for the pet insurance industry. Unlike human healthcare systems, veterinary medical records do not travel between clinics: there is no shared infrastructure to verify an animal’s diagnostic history across providers. Industry analyses estimate that claims fraud costs pet insurance managing general agents (MGAs) 3–7 points of loss ratio, translating to USD 300K–700K per USD 10M in written premium, while noting that even basic rules-based detection programs deliver a 3–10x return on investment [2]. As image-editing and generative tools become trivially accessible, fabricated or altered diagnostic imagery is a natural next step for claims fraud, mirroring the human-healthcare threat already demonstrated by Tordjman et al. [1].

Despite this, a review of the medical deepfake detection literature reveals that every existing dataset and method targets *human* imaging modalities (Section IV). An earlier proof-of-concept version of this work reported a dual-stream detector reaching 100% accuracy and AUC on a small (152-image) dataset. In preparing this expanded study, we audited that result and found a methodological confound: because the source photographs are downsampled from large, heterogeneous native resolutions while the DCGAN’s output is native 128×128 , the resize/construction history alone is separable, independent of any genuine forensic signal (Section VI-A). We treat this discovery, and its correction, as part of the paper’s contribution rather than something to quietly fix and hide: a benchmark that cannot be trusted to catch its own shortcuts is not a benchmark.

This paper makes the following contributions:

- 1) An audited, corrected experimental protocol with a genuine train/test separation at the generator level (Section VI), replacing an inflated proof-of-concept result with a defensible one.
- 2) Identification and legal verification of a second, substantially larger open veterinary radiograph dataset (VetXRay, 9,882 images, CC BY 4.0), in addition to the 152-image Mendeley VHS collection used previously (Section V).
- 3) A multi-generator synthesis and cross-generator generalization protocol (DCGAN, StyleGAN2-ADA, and a fine-tuned diffusion model), designed to test whether a detector learns *generator fingerprints* or genuine forensic signal (Sections VII–XII).
- 4) An explicit threat model separating *image authentic*-

ity detection from insurance fraud adjudication (Section III), positioning the detector as a forensic evidence signal within a broader anti-fraud pipeline rather than fraud proof in itself.

- 5) A public dataset, codebase, and reproducibility package to seed further research in this unexplored niche.

II. PROBLEM DEFINITION

The narrow version of our problem, “can a CNN plus a frequency descriptor separate a small DCGAN’s output from 152 real radiographs,” is not the question worth answering, and Section VI-A shows it can be answered affirmatively for reasons that have nothing to do with forensics. The question this benchmark targets instead is: *can a forensic detector reliably distinguish authentic veterinary radiographs from AI-generated or AI-manipulated radiographs under generator shift, common image transformations (compression, resizing, cropping), and previously unseen generative models?* This reframing is what elevates the work from a single-generator proof of concept to a benchmark, and it is why the experimental protocol in Sections XII–XVI exists.

III. THREAT MODEL

We formalize the setting a veterinary-radiograph forensic detector operates in, following the standard attacker/asset/capability framing used in media-forensics threat modeling.

Asset. The authenticity of a submitted diagnostic radiograph as evidence supporting an insurance claim (e.g., a fracture, a foreign-body ingestion, a cardiomegaly diagnosis).

Attacker. A claimant (or a party assisting a claimant) motivated to inflate, fabricate, or misattribute a veterinary diagnosis to obtain payment. We do not assume attacker sophistication in adversarial machine learning; we assume access to consumer-grade or open-source generative tools, and to standard image editors capable of resizing, compressing, and re-saving files.

Attack surface. (i) Fully synthetic radiograph submission, purporting to show a condition that never existed; (ii) locally manipulated radiographs, where a real study is altered to insert, remove, or exaggerate a finding; (iii) post-processing applied *after* generation or manipulation specifically to launder telltale artifacts (recompression, resizing, cropping, contrast/brightness adjustment).

Defender capabilities and knowledge assumptions. We evaluate under both a closed-world assumption (the defender has seen examples from the generator family used by the attacker; Section XI) and an open-world assumption (the attacker uses a generator family absent from the defender’s training data; Section XIII). We do not assume the defender can observe the attacker’s exact generation pipeline.

White-box vs. black-box. All experiments in this paper are black-box from the attacker’s perspective (the attacker has no access to the detector’s weights or gradients); a white-box, adaptive-attacker evaluation (an attacker who optimizes against a known detector) is explicitly out of scope and flagged as future work (Section XXVI).

Critical scope boundary. A positive detector output (“this image shows synthesis/manipulation artifacts”) is evidence relevant to a fraud investigation, not proof of fraudulent intent, and a negative output does not certify authenticity. The appropriate deployment is as one forensic signal feeding a broader claims-review pipeline that also considers claimant history, veterinarian attestation, and cross-provider record checks — the same layered posture used in image-based fraud detection outside insurance-specific image forensics [35]. We are not aware of a published, peer-reviewed threat-model paper specific to image-based insurance fraud; the closest published precedents address visual duplicate-claim detection rather than manipulation forensics [35], so this section is offered as an original contribution rather than a synthesis of prior work.

IV. RELATED WORK

A. Synthetic medical image generation

GANs and diffusion models are both used to synthesize medical images across modalities including MRI, CT, X-ray, ultrasound, and dermoscopy, motivated primarily by data augmentation and privacy-preserving data sharing [6]. Diffusion models have recently overtaken GANs in perceptual image quality, including for chest X-ray generation, but their clinical reliability and the artifacts they leave behind remain comparatively underexplored relative to GANs [6]. Foundational generator architectures relevant to this benchmark include DCGAN [12], StyleGAN2 [18] and its low-data variant StyleGAN2-ADA [19], DDPM [20], and latent diffusion models [21].

B. Detecting AI-generated medical images

MedForensics, presented at MICCAI 2025, spans six human modalities and twelve generative models, and proposes DSKI, a dual-stage knowledge-infusing detector built on a vision-language feature space [3]. M3Dsynth addresses AI-generated local manipulations in human 3D medical volumes [4]. Shahzad et al. train detectors on skin-cancer imagery against both GAN and diffusion generators, confirming that detectors calibrated to one generator family degrade sharply against the other [5] — precisely the phenomenon Section XII tests for veterinary imagery. Outside medical imaging, Wang et al. showed that a detector trained on a single GAN family (ProGAN) can generalize surprisingly well to unseen GANs under the right training recipe [22], while Ojha et al. propose universal, generator-agnostic fake-image detectors using frozen foundation-model features [23] — both directly inform our cross-generator protocol.

C. Frequency-domain and forensic feature analysis

A substantial body of work shows that CNN- and GAN-based generators leave periodic, checkerboard-like artifacts in the frequency domain, a direct consequence of transposed-convolution upsampling [7]. Frank et al. formalize this as a general frequency-analysis detection strategy [28]; Barni et al. use cross-band co-occurrence statistics [29]; Wolter et al. extend the idea to learned wavelet-packet representations [30];

Pontorno et al. specifically exploit DCT traces for generative-AI attribution [27]. Section XIV ablates spatial, FFT, DCT, and wavelet streams against each other for the first time on veterinary imagery.

D. Open-set and out-of-distribution detection

Because an insurer will eventually face a generator absent from any training set, we treat open-set detection as a first-class experiment (Section XIII), building on energy-based OOD scoring [24] and Mahalanobis-distance-based OOD/adversarial detection [25].

E. Explainability and shortcut learning

Grad-CAM [26] is the standard tool for localizing what a CNN attends to; we use it (Section XVIII) specifically to audit for shortcut learning, following the general methodological concern that a classifier can achieve high accuracy by exploiting dataset-construction artifacts rather than the phenomenon of interest — the exact failure mode we document in Section VI-A.

F. The veterinary gap

Existing machine-learning work on canine and feline radiographs, from groups at the University of Padua [14]–[17] and from Tahghighi et al. at the Ontario Veterinary College [8], [9], addresses classification of clinical findings and image quality, not synthetic-image generation or detection. The one veterinary paper we found addressing image *generation* uses a VAE to augment a small cardiomegaly dataset [34] — a data-augmentation goal, not a forensic one; we distinguish our detection-focused contribution from it explicitly. To our knowledge, no published dataset or benchmark addresses forensic detection of AI-generated or AI-manipulated veterinary diagnostic images.

V. VETERINARY IMAGING DATASET LANDSCAPE

Open, redistributable veterinary radiograph data is scarce. We surveyed Zenodo, Mendeley Data, Figshare, Kaggle, IEEE DataPort, TCIA, institutional repositories, and the data-availability statements of roughly twenty veterinary-radiology machine-learning papers published 2020–2026. Table I summarizes what we found.

The combination of Mendeley VHS and VetXRay yields approximately 10,034 real images under an unambiguous, redistribution-permitting license, an order of magnitude larger than any single dataset used in prior veterinary imaging ML work we could locate, and to our knowledge the first time these two collections have been combined for a machine-learning study. We verified VetXRay’s license directly against Zenodo’s machine-readable record API (`license.id = "cc-by-4.0"`, `access_right = "open"`) rather than relying on third-party aggregators, after finding that at least one independent inquiry into the same dataset had reported (incorrectly) that its license was unconfirmed. **[EXPERIMENT REQUIRED: confirmar compatibilidad de protocolo de adquisicion entre ambos datasets (vistas, resolucion) antes de combinarlos en un unico pool de entrenamiento; documentar si se usan combinados o por separado]**

Fig. 1. Representative real canine thoracic radiographs (top, blue border) from the Mendeley VHS collection [10] versus AI-generated counterparts from our DCGAN baseline (bottom, orange border). The generator’s outputs are visibly blurred and lack fine anatomical detail, consistent with the small-data regime discussed in Section VII.

Fig. 2. Both rows show the same underlying real radiographs; the only difference is resize history (Group A: direct downsample; Group B: reconstructed from an 8×8 seed). A detector separates the two groups at AUC = 1.000, demonstrating that resize/construction history alone is a strong, generator-independent confound.

VI. DATASET CONSTRUCTION

Figure 1 shows representative examples from both classes as they exist today, before the corrections described in this section: real canine thoracic radiographs from the Mendeley VHS collection [10] alongside DCGAN-generated counterparts. The visual gap is itself informative — it previews why Section VI-A finds the original 100% result unconvincing as forensic evidence: a generator this far from photorealism should not need frequency-domain forensics to be caught, and the fact that it appeared to require them is what motivated the audit below.

A. The resolution-history confound

Before any expansion, we audited the proof-of-concept pipeline and found that real images (large, heterogeneous native resolution, e.g. 1480×844 up to 2756×2800 pixels) are downsampled to 128×128 for the detector, while DCGAN outputs are natively 128×128 (built via five transposed-convolution upsampling layers from a random seed). We hypothesized that this asymmetry alone — not any genuine forensic signal — could explain a spuriously perfect detector, and confirmed it directly: a detector trained to distinguish two groups of images that are *both 100% real* (Figure 2), differing only in whether they were resized directly from native resolution (Group A) or reconstructed from an 8×8 seed mimicking a GAN’s upsampling construction (Group B), reaches AUC = 1.000, accuracy = 1.000 on a held-out test split ($n = 31$). A first, weaker version of this control (comparing single- vs. double-bicubic downsampling, without the seed-upsampling step) reached only chance-level AUC = 0.446 — the confound is specifically tied to the *upsample-from-a-tiny-seed* construction characteristic of GAN architectures, not to resizing in general. This confirms that a detector can reach a perfect score on this task without any exposure to a single generated image, and directly motivates the corrected protocol below.

B. Corrected split protocol

To address the confound and provide a genuine generalization test, we: (i) reserve a holdout set of 30 real images (of 152) that is never shown to the generator during training, training the DCGAN exclusively on the remaining 122 (unlike the original protocol, where the GAN’s discriminator saw 100% of the available real images); (ii) attempt a shared, randomized resize round-trip (uniform intermediate size in

TABLE I
VETERINARY RADIOGRAPH DATASET LANDSCAPE (SURVEYED 2026)

Dataset	Images	Species/region	Institution	License	Usable here?
Mendeley VHS [10], [11]	152–156	Canine, thoracic lateral	UABC, Mexico	CC BY 4.0	Yes (used previously)
VetXRay [13]	9,882	Canine & feline, thoracic	Univ. of Padua, Italy	CC BY 4.0	Yes (this paper)
Padua thoracic classification	3,839	Canine, thoracic	Univ. of Padua	Not public	No — request-only
OVC hemithorax/quality sets	899–900	Canine & feline, thoracic	Univ. of Guelph, Canada	Not public	No — request-only
VetAgro Sup feline set	500 (subset public)	Feline, thoracic	VetAgro Sup, France	Not public	No — request-only
DogHeart / Dog-Cardiomegaly	2,000	Canine, thoracic	Shanghai Aichong Pet Hospital	Undeclared, contradictory	No — unresolved

TABLE II
ORIGINAL VS. CORRECTED PROTOCOL (DCGAN ONLY); EXP-13
SHORTCUT AUDIT BEFORE AND AFTER THE ATTEMPTED MITIGATION

Metric	Original protocol	Corrected protocol
Accuracy	1.000	0.955
Precision	1.000	1.000
Recall	1.000	0.932
F1	1.000	0.965
Test AUC	1.000	1.000
n train/val/test	388/98/122	271/92/89
<i>EXP-13 shortcut audit (real-only, A vs. B)</i>		
AUC, no mitigation		1.000
AUC, with mitigation		1.000 (unchanged)

[16, 128], applied identically to both classes) before feature extraction, intended to prevent either class from carrying a cleaner or dirtier resize signature by construction; (iii) evaluate exclusively on the reserved real holdout (30 images) plus 59 synthetic images never used during detector training.

The proposed mitigation did not neutralize the confound. Re-running the EXP-00 audit *with* the shared round-trip applied still separates Group A from Group B at AUC = 1.000 (Table II), identical to the unmitigated control. We attribute this to an asymmetry the mitigation does not address: Group B’s information loss is fixed by an 8×8 bottleneck, more severe than any point in the [16, 128] range the shared round-trip draws from, so the shared step cannot retroactively erase a difference more extreme than its own range. We report this as a negative result rather than adjusting the range or hyperparameters post hoc to force a lower audit score, consistent with the pre-registered rule that experimental corrections are validated by the audit, not tuned against it.

Table II reports the resulting detector metrics under the corrected split. Accuracy drops from a perfect 100% to 95.5% (recall 93.2%, precision remains 100%), a genuine change driven at minimum by the newly enforced generator-level holdout, but because EXP-13 shows the resize-history shortcut persists, **we do not present 95.5% as confound-free forensic evidence**. The more decisive test of whether this detector captures genuine generative-model forensics, rather than DCGAN-specific construction artifacts, is cross-generator generalization (Section XII): a signal that transfers to StyleGAN2-ADA and diffusion-generated images, architectures with entirely different upsampling mechanics than DCGAN’s transposed convolutions, cannot be explained by this particular confound.

Fig. 3. Confusion matrix of the dual-stream detector under the corrected protocol (real holdout, $n = 30$; synthetic, $n = 59$). All 30 real holdout images are classified correctly; 4 of 59 synthetic images are misclassified as real, consistent with the 93.2% recall in Table II.

Fig. 4. VetRad-ForensicsNet: three complementary streams (spatial, frequency, noise-residual) combined via attention-based fusion before classification.

VII. SYNTHETIC RADIOGRAPH GENERATION

We synthesize radiographs from three generator families, chosen to span the practical range of what an attacker could plausibly use: (i) DCGAN [12], already trained as a low-data baseline; (ii) StyleGAN2-ADA [19], NVIDIA’s adaptive-discriminator-augmentation method designed specifically for GAN training in the low-to-moderate data regime our combined dataset occupies; (iii) a diffusion model [20], [21] fine-tuned via low-rank adaptation on the same real image pool. Training for (ii) and (iii) uses Kaggle Notebooks’ free GPU tier (NVIDIA P100 or T4 $\times 2$, reset weekly), with paid cloud compute as a fallback if the free quota is insufficient. **[EXPERIMENT REQUIRED: resultados de entrenamiento de StyleGAN2-ADA y del modelo de difusion: FID/KID vs. real, ejemplos generados]**

VIII. PROPOSED ARCHITECTURE: VETRAD-FORENSICSNET

Building on the dual-stream design of the proof of concept (a ResNet-18 spatial branch plus a radially-averaged FFT frequency branch, fused before a classification head), we extend the architecture with a third, noise-residual stream, and an attention-based fusion layer in place of simple concatenation (Figure 4). The three streams target complementary evidence: anatomical/spatial plausibility (spatial stream), periodic generator fingerprints (frequency stream, generalized in Section XIV to FFT, DCT, and wavelet-packet variants), and sensor/generation noise statistics (residual stream, computed as the difference between an image and a denoised version of itself). Section XIV already reports a pilot four-way spatial+FFT+DCT+wavelet fusion (Table III); the noise-residual stream and the full five-configuration ablation (spatial; spatial+frequency; spatial+noise; frequency+noise; spatial+frequency+noise) are implemented per Section X but not yet run to completion.

[EXPERIMENT REQUIRED: resultados del ablation completo de 5 configuraciones de streams, incluyendo el residual de ruido]

IX. EXPERIMENTAL PROTOCOL

[EXPERIMENT REQUIRED: describir protocolo final una vez fijado: splits, semillas, numero de repeticiones (5-10), hiperparametros por experimento]

X. BASELINE DETECTOR ARCHITECTURES

In addition to the ResNet-18 dual-stream baseline, we evaluate ResNet-50, EfficientNet-B0, ConvNeXt-Tiny, and a Vision Transformer as spatial backbones, chosen to span convolutional and transformer-based architectures at comparable parameter budgets, and to test whether the frequency-branch fusion strategy is backbone-agnostic. Because torchvision does not ship a ViT-Small checkpoint, we substitute ViT-B/16 (ImageNet-1K weights) explicitly and flag every result derived from it accordingly, rather than silently approximating a different model. In a 2-epoch pilot verification against DCGAN alone, ResNet-18 (the architecture used throughout this paper) reaches 0.978 accuracy and 0.998 ROC-AUC, consistent with Table II’s longer 25-epoch run; the remaining four backbones are implemented and unit-tested for correct output shape and gradient flow, but not yet trained end-to-end pending the full multi-generator dataset. [EXPERIMENT REQUIRED: tabla comparativa completa de las 5 arquitecturas entrenadas 25 epocas bajo los 3 generadores]

XI. IN-DISTRIBUTION DETECTION

For DCGAN, in-distribution detection under the corrected split protocol is reported in Table II (Section VI): 95.5% accuracy, 100% precision, 93.2% recall, with the caveat that the resize-history shortcut audited in Section VI-A is not confirmed neutralized, so this number should be read as an upper bound pending the mitigation work discussed there. [EXPERIMENT REQUIRED: resultados equivalentes para StyleGAN2-ADA y difusion una vez entrenados]

XII. CROSS-GENERATOR GENERALIZATION

The central question of this benchmark is whether a detector trained on one generator family transfers to another, or instead memorizes generator-specific fingerprints. We construct a train-generator $\times$ test-generator matrix across DCGAN, StyleGAN2-ADA, and the fine-tuned diffusion model (including training on two families and testing on the third, held out entirely), reporting accuracy, precision, recall, specificity, F1, ROC-AUC, PR-AUC, Matthews correlation coefficient, and balanced accuracy for every cell. The evaluation pipeline for this matrix is implemented and verified end-to-end with the one generator family available at the time of writing (DCGAN, diagonal cell only: accuracy 0.978, ROC-AUC 0.998, balanced accuracy 0.967, matching Table II); it explicitly reports which generators are unavailable rather than fabricating off-diagonal cells. [EXPERIMENT REQUIRED: matriz cross-generator 3x3 completa, con mapa de calor, una vez disponibles StyleGAN2-ADA y difusion]

TABLE III
FREQUENCY-DOMAIN ABLATION, PILOT RUN (2 EPOCHS, DCGAN ONLY)

Features	Acc.	Spec.	F1	ROC-AUC	Bal. Acc.
Spatial only	1.000	1.000	1.000	1.000	1.000
FFT only	0.663	0.000	0.797	0.383	0.500
DCT only	0.663	0.000	0.797	0.722	0.500
Wavelet only	0.663	0.000	0.797	0.393	0.500
Spatial+FFT+DCT+Wavelet	0.989	0.967	0.992	1.000	0.983

XIII. OPEN-SET DETECTION

We evaluate whether energy-based [24] and Mahalanobis-distance-based [25] out-of-distribution scoring, plus a one-class baseline, can flag a generator family entirely absent from training as “unknown synthetic” rather than misclassifying it as real. [EXPERIMENT REQUIRED: resultados de deteccion open-set con un generador oculto]

XIV. FREQUENCY, DCT, AND WAVELET FORENSICS

We ablate the forensic feature space across spatial-only, FFT-only [7], [28], DCT-only [27], wavelet-packet-only [30], and a fused combination, to determine which transform domain contributes discriminative signal for veterinary radiographs specifically, and whether the answer differs by generator family.

Table III reports a pilot verification run (2 epochs, DCGAN only, $n = 89$ test images; the full protocol runs 25 epochs across all three generator families once StyleGAN2-ADA and diffusion outputs are available). Even at this pilot scale, the pattern is informative: at the fixed 0.5 decision threshold, FFT-only, DCT-only, and wavelet-only branches each classify every test image as synthetic (specificity = 0, balanced accuracy = 0.5), yet their ROC-AUC values differ meaningfully (0.38–0.72), showing each transform domain carries some ranking signal despite an uncalibrated threshold after only 2 epochs. The spatial-only branch alone already reaches a perfect score, corroborating Section VI-A: with the resolution-history confound still active, spatial appearance alone is sufficient to solve this particular pilot, independent of any frequency-domain analysis. The fused (all) configuration recovers a strong result (accuracy 0.989, balanced accuracy 0.983) without matching spatial-only’s ceiling, consistent with the fusion head needing more than 2 epochs to fully exploit the added features.

[EXPERIMENT REQUIRED: repetir esta tabla con 25 epocas y los 3 generadores una vez disponibles StyleGAN2-ADA y difusion]

XV. ABLATION STUDIES

[EXPERIMENT REQUIRED: ablation de VetRad-ForensicsNet (5 configuraciones de streams) y ablation de dominio de frecuencia (Seccion XIV) consolidados]

XVI. ROBUSTNESS AND ANTI-FORENSICS

An attacker is unlikely to submit a generator’s raw output unmodified. We evaluate detector performance after JPEG compression (quality 10–95), resizing, cropping,

Gaussian blur, Gaussian noise, contrast/brightness/gamma adjustment, sharpening, and repeated recompression, reporting performance-vs-severity curves for the most damaging transformations. [EXPERIMENT REQUIRED: *curvas de robustez: accuracy/AUC vs. calidad JPEG, resize, blur, ruido*]

XVII. GENERATOR ATTRIBUTION

Beyond binary real/fake detection, we evaluate a multiclass task (real vs. each individual generator family) to test whether generator-specific fingerprints are identifiable, and whether they survive the robustness transformations of Section XVI. [EXPERIMENT REQUIRED: *resultados de atribucion multi-clase y su degradacion bajo post-procesamiento*]

XVIII. EXPLAINABILITY

We apply Grad-CAM [26] to the spatial stream and a frequency-domain saliency map to the frequency stream, to visualize whether the detector attends to anatomically plausible regions, edges, radiographic markers, or background/artifact regions — direct evidence for or against the shortcut-learning failure mode documented in Section VI-A. [EXPERIMENT REQUIRED: *galeria de mapas Grad-CAM y de frecuencia por generador*]

XIX. FAILURE ANALYSIS

[EXPERIMENT REQUIRED: *casos de falsos positivos/negativos de las Secciones XII, XVI, y XIII, con ejemplos visuales y analisis*]

XX. STATISTICAL ANALYSIS

All headline comparisons are reported as mean $\pm$ standard deviation across 5–10 independent seeds, with 95% bootstrap confidence intervals, rather than a single point estimate. We distinguish three comparison types, each with a matched test: (i) comparing two ROC curves from the *same* test set (e.g., ResNet-18 vs. ResNet-50 dual-stream detectors evaluated on identical held-out images) uses the DeLong test for correlated AUCs [31], which accounts for the correlation induced by sharing test examples rather than treating the two AUCs as independent; (ii) comparing paired per-example correctness between two classifiers on the same test set (e.g., spatial-only vs. fused features) uses McNemar’s test on the discordant-pair count [32]; (iii) confidence intervals on a single metric (accuracy, AUC, or a cross-generator matrix cell) use the non-parametric percentile bootstrap [33] with 10,000 resamples of the test set. Every JSON result file already stores per-example probabilities and labels (Section XXIV), which is what these three procedures consume; no additional data collection is required to apply them once the underlying experiments (Sections XII–XVII) are complete. [EXPERIMENT REQUIRED: *resultados estadisticos finales: intervalos de confianza, p-valores DeLong/McNemar para las comparaciones principales*]

XXI. INSURANCE-FRAUD DEPLOYMENT SCENARIO

Consistent with the threat model of Section III, we discuss how a detector of this kind would sit within a claims-review pipeline: as a triage signal that routes suspicious submissions to human review, not as an automated denial mechanism, and with explicit logging of confidence and the specific forensic evidence (frequency anomaly, spatial implausibility, or both) driving each flag.

XXII. ETHICAL AND LEGAL CONSIDERATIONS

All real imagery used in this work is licensed CC BY 4.0 with redistribution and derivative-work rights explicitly confirmed against primary sources (Mendeley Data’s public API for [10], Zenodo’s record API for [13]), and both datasets are de-identified at the patient level by their original publishers. We did not use, and explicitly excluded, any dataset whose licensing terms were unclear, request-only, or contradictory (Section V). The human Visual Turing Test design described in Section XXVI was not executed in this study; it would require institutional ethics review and informed consent, which are prerequisites we did not pursue here.

XXIII. THREATS TO VALIDITY

Internal validity. Section VI-A documents a confound we found in our own initial protocol and an attempted mitigation that, audited empirically, did *not* neutralize it — we report this as a negative result rather than retroactively tuning the mitigation to pass its own audit. We cannot rule out that other, undiscovered dataset-construction artifacts remain, and the single-generator (DCGAN) accuracy figures in Section XI should not be read as confound-free until the cross-generator test of Section XII is complete. **Construct validity.** Detecting a specific generator’s output is not the same as detecting all future generators; Section XIII probes but does not solve this. **External validity.** All real data covers thoracic radiographs in dogs and cats; generalization to other views, species, or modalities (ultrasound, dental radiographs) is untested. **Statistical validity.** With approximately 10,000 real images the dataset is far larger than the original 152-image proof of concept, but still modest relative to human-imaging forensics benchmarks; Section XX’s confidence intervals should be read accordingly.

XXIV. REPRODUCIBILITY

All code, trained weights, dataset splits, and configuration files accompany this paper in a public GitHub repository (jmmana/VetScanForensics) and public Kaggle datasets. Local (Apple M4 Pro, PyTorch Metal Performance Shaders) and cloud (Kaggle Notebooks, NVIDIA P100/T4 $\times$ 2) runs use Python 3.14 with PyTorch 2.12.0 and torchvision 0.27.0; StyleGAN2-ADA and diffusion training run under Python 3.12 in Kaggle’s managed environment with stylegan2_pytorch and diffusers, respectively. All detector and audit scripts fix the random seed to 42 for split generation, weight initialization, and evaluation-time preprocessing (Section VI); training-time data augmentation uses PyTorch’s default per-epoch RNG state, documented per run

in each result JSON alongside library versions, an exact train/val/test file manifest, and a hash of the real-image split file, so a given result can be traced back to the precise data partition that produced it. Every experiment script writes a machine-readable JSON result file (probabilities, labels, confusion matrix, and configuration) rather than only a summary metric, so post hoc re-analysis (e.g., recomputing a metric we did not originally report) does not require re-running the model. **[EXPERIMENT REQUIRED: confirmar hashes de dataset finales y checkpoints publicados tras la fase de Kaggle]**

XXV. DISCUSSION

[EXPERIMENT REQUIRED: síntesis final una vez completados los experimentos de las Secciones XII–XVII]

XXVI. FUTURE WORK

Beyond StyleGAN3 and additional diffusion variants, the most direct extension is a real Visual Turing Test comparing human (veterinarian, veterinary student, and lay) detection accuracy against the detector on a held-out set of 20–50 radiographs — designed here (Section XXII) but explicitly not executed pending institutional ethics approval and informed consent. Extending the real-image base to additional views (orthopedic, abdominal) and to feline-specific pathology beyond VetXRay’s existing feline component is a second natural direction, contingent on identifying further openly licensed data (Section V).

XXVII. CONCLUSION

We identified a real, unaddressed gap between two active but previously disconnected lines of work: the growing literature on detecting AI-generated human medical images, and the well-documented, costly problem of veterinary insurance claims fraud. In building this benchmark, we found and corrected a methodological confound in our own initial result, identified a second large open dataset that expands the field’s data landscape by an order of magnitude, and laid out a multi-generator, cross-generalization, robustness, and explainability evaluation protocol intended to withstand scrutiny that a single-generator proof of concept could not. **[EXPERIMENT REQUIRED: cerrar con los resultados finales una vez completada la ejecución]**

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